

1. Administrative & Study Identification

Full Study Title: A Phase 2a Randomized, Placebo-Controlled Clinical Study to Evaluate the Efficacy and Safety of MK-5475 in Adults With Pulmonary Hypertension Associated With Chronic Obstructive Pulmonary Disease **Short Title/Acronym:** INSIGNIA-PH-COPD / Frespaciguat (MK-5475) **Protocol Number:** MK-5475-013 (Other IDs: 2022-501201-13-00) **NCT Number:** NCT05612035 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** MK-5475 (Frespaciguat) / Inhaled soluble Guanylate Cyclase (sGC) stimulator **Phase of Development:** Phase 2 (Phase 2a) **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * Primary Objective: To evaluate whether MK-5475 (frespaciguat) is superior to placebo in increasing the 6-Minute Walking Distance (6MWD) from baseline at Week 24. *

Secondary Objectives: To evaluate the overall safety and tolerability of MK-5475 and observe changes in WHO Functional Class and pulmonary hypertension symptoms.

2.2 Background & Rationale Pulmonary hypertension (PH) is a strong predictor of mortality in COPD patients, characterized by high blood pressure in the pulmonary arteries, which overworks the heart and limits physical activity. Traditional oral blood pressure medications can cause systemic hypotension (low blood pressure) and hypoxemia (low oxygen levels in the body). MK-5475 is a novel study medication delivered directly to the lungs via an inhaler. It is designed to selectively lower blood pressure in the pulmonary arteries without causing systemic low blood pressure or reducing oxygen levels in the rest of the body.

3. Study Design & Methodology

3.1 Design Framework * Study Type: Interventional * **Control Type:** Placebo-controlled * **Blinding:** Double-blind * **Randomization:** Randomized

3.2 Planned Enrollment & Duration * Target \$N\$: Approximately 120 evaluable patients * **Number of Sites:** Global (Including US, UK, and 18 other locations) * **Estimated Study Start:** 15-Mar-2023 * **Estimated Primary Completion:** 07-Apr-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 40 to 85 years inclusive. 2. Has Group 3.1 pulmonary hypertension associated with chronic obstructive pulmonary disease (PH-COPD) with a right heart catheterization (RHC) at screening or within 12 months before screening meeting hemodynamic criteria. 3. Has a physician diagnosis of obstructive lung disease on pulmonary function testing (PFT) and a WHO Functional Class assessment of Class II to IV.

4.2 Exclusion Criteria 1. Has Group 1, 2, 4, or 5 pulmonary hypertension, or non-COPD related Group 3 PH. 2. Has significant left heart disease, or has had acute coronary syndrome, coronary artery bypass graft, or percutaneous coronary intervention within 2 months before randomization. 3. Has used PAH-specific therapies or initiated a pulmonary rehabilitation program within 2 months of randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** 380 µg of frespaciqualat (MK-5475) or matching placebo once daily. * **Route of Administration:** Oral inhalation via inhaler device. * **Duration of Treatment:** 24 weeks for the base period, followed by an optional extension period of up to 42 months.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Stable and optimized chronic COPD-specific baseline therapy; stable supplemental oxygen; stable use of PDE5 inhibitors, antihypertensives, diuretics, or anticoagulants (initiated at least 3 months prior). * **Prohibited:** Initiation of new PAH-specific therapies or new pulmonary rehabilitation programs during the 24-week base period.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Day -42 to 0	Informed Consent, Medical History, PFT, RHC (if no historical data), 6MWT Baseline
Baseline	Day 0	Randomization, First Inhaler Dose, Device technique training
Interim	Weeks 1 - 24	Regular Onsite Visits: 6MWT Efficacy Assessments, Safety & AE tracking
End of Study	Week 24 / Ext.	Primary endpoint evaluation at 24 weeks; Entry to up to 42-month extension; Final phone follow-up 2 weeks after the last visit

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Change from baseline in 6-Minute Walking Distance (6MWD) at Week 24. * **Measurement Method:** Standardized 6-Minute Walk Test (6MWT) measured at clinical sites.

7.2 Secondary & Exploratory Endpoints * Safety and tolerability profile (incidence of Adverse Events/Serious Adverse Events). * Change from baseline in WHO Functional Class.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Standard collection via regular clinical visits and a mandatory telephone follow-up visit 2 weeks after the last study visit.
 - **Serious Adverse Events (SAEs):** Routine reporting within 24 hours of investigator awareness.
 - **Data Monitoring Committee (DMC):** Utilized for ongoing safety signal detection (Standard sponsor oversight).
-

9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy (6MWD change) and Safety Set for adverse events.
 - **Sample Size Justification:** $N=120$ powered to detect a clinically meaningful difference in 6MWD at Week 24 between the frespaciguat 380 µg and placebo arms.
 - **Handling of Missing Data:** Standard Multiple Imputation (MI) techniques for continuous endpoints like 6MWD dropouts.
-

10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine on-site and remote monitoring visits to ensure 6MWT standardization and protocol adherence.
 - **Audit Trail:** Standard electronic Case Report Form (eCRF) tracking, locked query resolution, and data clarification forms.
-

11. Study Identifiers & Governance

- **Primary ID:** MK-5475-013
- **Registry Number:** NCT05612035
- **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC
- **Study Title:** INSIGNIA-PH-COPD

- **Official Regulatory Title:** A Phase 2a Randomized, Placebo-Controlled Clinical Study to Evaluate the Efficacy and Safety of MK-5475 in Adults With Pulmonary Hypertension Associated With Chronic Obstructive Pulmonary Disease
-

12. Clinical Framework (PH-COPD Specific)

- **Primary Condition:** Pulmonary Hypertension associated with Chronic Obstructive Pulmonary Disease (PH-COPD)
 - **Diagnosis Threshold:** Group 3.1 PH-COPD confirmed via Right Heart Catheterization (RHC) and WHO Functional Class II-IV
 - **Medical Methodology:** Inhaled soluble Guanylate Cyclase (sGC) stimulator combined with stable baseline COPD maintenance therapy
 - **Optimization Target:** Improvement in exercise capacity measured by 6MWD
-

13. Study Procedures & Methodology

- **Management Pathway:** Regular monitoring of 6MWD and WHO Functional class symptom progression
 - **Education Protocol (HCP):** Standardization training on conducting the 6-Minute Walk Test (6MWT) across all global sites
 - **Education Protocol (Patient):** Proper dry powder inhaler (DPI) device technique training and dosing adherence
 - **Quality Audit Mechanism:** Centralized safety monitoring and AE follow-up tracking
-

14. Observation & Follow-Up Schedule

- **Total Duration:** 24-week base period + optional extension up to 42 months
 - **Onsite Visit Cadence:** Frequent clinic visits (approx. 30 visits over a potential 43-month maximum participation)
 - **Remote Monitoring Frequency:** Telephone follow-up scheduled 2 weeks following the last study visit
 - **Checklist Review Interval:** Continuous AE and concomitant medication review throughout the 24-week base observation phase
-

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				